BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2018

CHEMISTRY

Past Paper 2018 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

SECTION - C (MULTIPLE CHOICE QUESTIONS)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q. 16: Choose the correct answer from the given options and write the same in your answer copy. Note: Cutting and Over writing is not allowed.

Q1. The pair of isoelectronic ions is which one?

(A) H^- and Li
(B) Ba^{2+} and Cs^+
(C) N^{-3} and Ne
(D) None of these

Q2. Which one of the following is hydride but not binary compound?

(A) HOBr
(B) H_2SO_4
(C) $Fe(OH)_3$
(D) All of these

Q3. The product obtained at Solvay tower in ammonia Solvay process is:

(A) $CaO+CO_2$
(B) $NH_4Cl + NaHCO_3$
(C) NH_3+O_2
(D) Amonical brine

Q4. None of the following represents isomers except:

(A) $C_2H_5OCH_3$; C_2H_6O
(B) $CH_3COC_2H_5$; CH_3-CH_2-CHO
(C) $(CH_3)_2CH$-CH_2-CH_2 [with OH group] ; CH_2–$CH(CH_3)CH_2(CH_3)$ [with OH group]
(D) $CH_2 = CH (C_2H_5)$; $CH \equiv C (C_2H_5)$

Q5. 2 – chloro butane reacts with sodium ethoxide to produce:

(A) Isobutoxy ethane
(B) 2 – ethoxy butane
(C) 2 – ethyl butane
(D) Isohexane

Q6. The complete combustion of 1 mole methane needs 2 moles O_2 (oxygen) the same for C_2H_2 will be:

(A) 2.5 moles
(B) 3.5 moles
(C) 1.5 mole
(D) 3 moles

Q7. The mixture in water consisting 4% CH_3OH & 85% C_2H_5OH is called which one?

(A) Absolute alcohol
(B) Rectified spirit
(C) Methylated spirit
(D) All of these

Q8. The chemical composition of Gypsum salt is:

(A) KNO_3
(B) $MgSO_4.7H_2O$
(C) $CaSO_4.2H_2O$
(D) Na_2[illegible in source scan]

Q9. The isotopes of elements (Hydrogen) differ to one another by:

(A) No. of neutrons
(B) No of protons
(C) No. of electrons
(D) Atomic number

Q10. Among the following metals which one can be corroded:

(A) Iron
(B) Potassium
(C) Silver
(D) Tin

Q11. The basic amino acid is which one?

(A) Lysine
(B) Alanine
(C) Glutamic acid
(D) Isoleucine

Q12. The corrosive and hygroscopic is none of these except:

(A) HNO_3
(B) H_3BO_3
(C) H_2SO_4
(D) Cl_2

Q13. The reducing agent used in silvering of mirror for Ag^+ to Ag is which one?

(A) NH_4OH
(B) $CaCO_3$
(C) AgBr
(D) $C_6H_{12}O_6$

Q14. The triad representing the regular increase in radii(size) is:

(A) Ca, Sr, Ba
(B) I^-, Cl^-, Br^-
(C) Ga, C, Si
(D) All of these

Q15. The material used to make designs on the glass is:

(A) Cupric oxide
(B) Calcium carbonate
(C) Hydro fluorine acid
(D) Boron oxide

Q16. Markownikoff's rule is applicable for addition reaction of which one?

(A) But-ene
(B) Propyne
(C) Vinyl bromide
(D) All of these

Q17. The valeric acid will be the product obtained by reacting "CO_2" in water with: [scan truncated below the options in the source]

(A) R-Mg-x
(B) C_4H_9-Mg-I
(C) C_2H_5-Mg-Cl
(D) C_5H_{11}-Mg-Br

SECTION – A (SHORT ANSWER QUESTIONS)

Time Allowed: 2 Hours and 40 minutes | **Maximum Marks:** 40

Note: NOTE: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. Explain why? i) Ionization potential increases in period. ii) H_2S is reducing agent but H_2SO_4 is oxidizing agent being both contain Sulphur. iii) Graphite is good conductor but diamond is bad conductor being both contain carbon.

Q2. Define structure; compare the structural arrangement of Ethane and Ethene.

Q3. How would you prepare sodium metal (Na) from "Brine" by electrolysis in Down's cell & give its diagram.

Q4. Give the chemical behaviour's in following with balanced equation. (i) Aluminum (Al) with moist air (O_2) and show the nature of reaction (ii) Lead chromate ($PbCrO_4$) digested in NaOH. (iii) Saturated solution of Na_2CO_3 with CO_2 in water.

Q5. Explain the following terms with reaction: 1) Sponification of oil and fat. 2) Phosphatic fertilizers

Q6. Define with examples the following: i) Periods and groups in modern periodic table. ii) Colour of the compounds of transition elements iii) Ligands in complex molecules.

Q7. Benzene gives electrophilic substitution(SE) reactions but not nucleophilic (SN) why? Explain its nitration reaction with mechanism.

Q8. Change the following names into IUPAC system: (i) Methyl ,Isopropyl ethylene (ii) Aspartic acid (iii) Ethyl, terbutyl ether (iv) Neohexyl alcohol (v) Vinyl acetate

Q9. Hydrogen is placed over the periodic table why? Differentiate the atomic and molecular existence of Hydrogen.

Q10. Give the preparation of ethyl bromide (C_2H_5Br) from: (i) Ethane (ii) Ethyl alcohol (iii) Ethene and mention the type reaction of each change.

Q11. (a) Correct the following incorrect names. (i) 3- isopropyl butanal (ii) 2- ethyl, Hexan-4-one (iii) N-ethyl, methanamine (b) Give the general formula used to prepare homologous series of Alkynes and Ethers.

Q12. What is bleaching powder? Give the preparation and properties of Epsom salt.

SECTION – B (LONG ANSWER QUESTIONS)

Maximum Marks: 28

> Note: NOTE: ATTEMPT ANY TWO (02) QUESTIONS.

Q13.

(a) [04] Give the reaction of acetylene with (i) Hot $KMnO_4$ (ii) Water in H_2SO_4/ $HgSO_4$ + heat

(b) [04] Describe the destructive distillation of coal.

(c) [06] Explain the product formed by reacting formaldehyde with. (i) Phenyl hydrazine (ii) NaOH (iii) H_2 in Na/ C_2H_5OH

Q14.

(a) [04] Write detailed note on Photography

(b) [06] Explain by balance chemical equations the reactions of Cl_2 gas with. i) NaOH ii) C_2H_4 iii) $Ca(OH)_2$

(c) [04] Distinguish between crystalline & amorphous sulphur

Q15.

(a) [06] What are carbohydrates, give their classification

(b) [04] Give properties & preparation of 6, 6I – Nylon.

(c) [04] Define simple and conjugate proteins with examples